

0-9

gate 2025, CS6, DA
algorithm

→ Asymptotic notation

Lec 10

Lec_9

Practice Questions on time Complexity

1-24

Time complexity = How many times
loop is getting
executed

$$\underbrace{l = l^2}$$

25.) Main()

```
{  
  for(i=1; i^2 <= n; i++)  
  {  
    x = y + z;  
  }  
}
```

loop $\sqrt{n}$
times

$$O(\sqrt{n})$$

$$\begin{aligned} n^2 &= n \\ n &= \sqrt{n} \end{aligned}$$

$i=1$, $\checkmark 1^2 \leq n$, $i++$
 $i=2$, $\checkmark 2^2 \leq n$, $i++$
 $i=3$, $\checkmark 3^2 \leq n$, $i++$
 $i=4$, $\checkmark 4^2 \leq n$
:
 $i=n$, $n^2 > n$

fail

$$1^2 < n$$

$$2^2 < n$$

$$3^2 < n$$

$$\vdots$$

$$h^2 < n$$

$$\underline{n^2 > n}$$

$$n^2 = n$$

$$n = \sqrt{n}$$

✓
26.) Main()

```
{  
  for(i=1; i<=n; i++)  
  {  
    for(j=1; j<=i; j++)  
    {  
      x=y+z;  
    }  
  }  
}
```

→ 4
→ 2

outer loop

$i=1$, $i<=n$, $i++$

$i=1$, $j<=1$, $j++$
 $j=2$, $2<=1$, 1 times

$i=2$, $2<=n$, $i++$

$i=1$, $j<=2$, $j++$
2 times

$i=3$, $3<=n$, $i++$

3 times $j=1$, $j<=3$

for (i=1; i<=n; i++)

{ for (j=1; j<=i; j++)

{ n = y + z

}

}

$$n = \frac{n^2 + n}{2}$$

i=1,
i=2,
i=3,
i=4,
i=5

1 times
2 times
3 times
4 times
5 times

1 + 2 + 3 + 4 + ... + n

$$\Rightarrow \frac{n(n+1)}{2} = \frac{n^2 + n}{2}$$

$$O\left(\frac{n^2 + n}{2}\right) = O(n^2)$$

$O(n^2)$

for($i=1$; $i \leq n$; $i++$)

{ for($j=1$; $j \leq n$; $j++$)

{ $n = i + j$

}

$n + n + n + \dots$
 $n[1 + 1 + 1 + \dots + n \text{ times}]$

$n \times n$
 $= \underline{O(n^2)}$

$i=1$,
 $i=2$,
 $i=3$,
 $i=4$,
 $i=n$

n times
 n times
"
"
"
"

27.) Main()

```

{
  for(i=1; i<=logn; i++)
  {
    for(j=1; j<=2^i; j++)
    {
      x=y+z;
    }
  }
}

```

$i=1,$

$j=1, j<=2^1, j++$
2 times.

$\frac{1=1}{2^1},$ 2 times

$i=2$

$j=1, j<=2^2, j++$

$\frac{1=2}{2^2},$ 4 times.

$i=3$

$j=1, j<=2^3, j++$

$\frac{1=3}{2^3},$ 8 times.

$i=1, 2, 3, 4, \dots, \log n$

$\frac{1}{2^1} + \frac{2}{2^2} + \frac{3}{2^3} + \frac{4}{2^4} + \dots + \frac{\log n}{2^{\log n}}$

$\Rightarrow$

$$r = \frac{2^2}{2^1} = 2$$

$$2^1 + 2^2 + 2^3 + \dots + 2^{\log_2 n}$$

n terms in G.P. $\rightarrow$ first term

$$\text{sum} = \frac{a(r^n - 1)}{r - 1}$$

$$= \frac{2(2^{\log_2 n} - 1)}{2 - 1}$$

$$O(2^{n-2})$$

$$= \underline{\underline{O(n)}}$$

$$= 2(2^{\log_2 n} - 1)$$

$$= 2(n^{\log_2 2} - 1)$$

$$= 2(n^1 - 1)$$

$$= 2(n - 1)$$

$$= 2n - 2$$

Let n is a prime number.

28.) Main()

for($i=2; i \leq n/2; i++$)

{ if($n \% i == 0$)

{ for($j=1; j \leq 2^i; j++$)

{
x=y+z;
}

always

i, n whole no.
 n is prime number

13

$n=13$

13

1 to $n = n$ hms
2 to $n = n-1$
3 to $n = n-2$
($\frac{n}{2} - 2$)

for($i=2; i \leq 7.5; i++$)

$13 / 2 \neq 0$
 $13 / 3 \neq 0$

$13 / 4 = 0$

$\frac{n-2}{2}$

$n/2$

